HeatNet Member 3: Viva Notes (Data, Compare and Analysis)

A short explanation

"I built HeatNet's data layer. It stores sessions and readings locally with Room, lets users reopen, rename and delete sessions, finds weak spots in a room, and compares two sessions to show whether a change helped. Comparison only uses room areas that both sessions measured, and it warns when the comparison may be unfair."

What I built

Piece	What it does	Where
Local storage	Saves sessions (outline, size, connection type) and readings (position, metrics, loss method, radio fields, and status) in Room. A legacy marker identifies simulated readings saved by older versions; current app sessions create real measurements only. No server or account.	`data/.../db/`
Repository	One entry point for the app: create session, add and delete readings, load, rename, delete.	`SessionRepository.kt`
Session list	Saved sessions, with rename, delete and compare picking.	`app/.../SessionListScreen.kt`
Weak-spot detection (F8)	Flags problem corners of the room and writes one sentence per corner.	`WeakSpotDetector.kt`
Session comparison (F7)	Before and after numbers per metric, a verdict, and warnings.	`SessionComparison.kt`, `CompareScreen.kt`

How weak-spot detection works

1. It needs at least 3 readings. With fewer it asks for more.
2. It cuts the room's bounding box into four quadrants around its centre (north-west, north-east, south-west, south-east).
3. A quadrant is flagged for a problem when at least half of its readings that have that metric show the problem.
4. Starting thresholds (from the PRD, to be tuned after real-room testing):

Problem	Flagged when
Weak signal (Wi-Fi)	RSSI below -75 dBm
Weak signal (mobile)	Below -100 dBm (an assumption, not yet tuned on our phones)
High latency	Above 100 ms
Packet loss	Above 2%, ICMP only
Slow download	Below 25% of the best download in that session

5. The output is one sentence per flagged quadrant, for example "Weak signal and high latency in the north-east corner".
6. Failed tests (null values) are ignored for that metric only. HTTP-failure rates are ignored for packet loss, because they are not real packet loss.

How comparison works

- Only sessions on the same connection type can be compared. Two Wi-Fi sessions, or two mobile sessions.
- Both sessions need at least one reading. The same session cannot be compared with itself.
- Each room is divided into a 4×4 grid. Only cells that both sessions sampled are compared. This keeps the comparison fair when readings were taken in different places.
- For each metric it gives the before mean, the after mean and the difference.
- The verdict is IMPROVED, WORSE, SIMILAR or NOT_ENOUGH_DATA. A difference within 5% counts as SIMILAR (noise). Direction respects whether higher or lower is better for that metric.
- Both maps use one shared colour range, so a small change does not look big.
- Warnings appear when:

- the room sizes differ by more than 20%,
- the sessions used different access points (BSSID),
- the sessions have no shared sampled areas,
- any reading is legacy simulated data from an older app version,
- HTTP request-failure rates are present (they are not compared as ICMP loss).

Likely questions

1. What did you personally build? Storage with Room, the repository, the session list with rename and delete, weak-spot detection and session comparison.

2. Why store positions in metres? Pixels depend on screen size. Metres keep saved sessions and comparisons valid on any device.

3. Why local storage and no backend? The PRD rules out accounts, login and cloud sync. It keeps the project simple and keeps data on the phone.

4. How does weak-spot detection decide a corner is bad? At least half of that corner's readings (those with the metric) cross a threshold.

5. Why quadrants? It gives the user a plain-language answer ("north-east corner") without needing a floor plan.

6. Why is packet loss ICMP-only in your analysis? The HTTP fallback counts failed web requests, which is an application-level signal, not dropped packets. Mixing them would be misleading.

7. How do you know your thresholds are right? We do not yet. They are the PRD's starting values. The plan is to tune them after real-room testing, and the mobile-signal threshold is an assumption.

8. Why only compare shared areas? If one session sampled the kitchen and the other the bedroom, comparing their averages says nothing about a change. Shared grid cells make it a like-for-like comparison.

9. Why a 5% "similar" band? Network measurements are noisy. Small differences should not be reported as improvements.

10. Why must both sessions have the same connection type? Wi-Fi and mobile data use different technologies and signal scales, so their numbers are not comparable.

11. What is `isDemo`? It is a compatibility flag for simulated readings saved by older app versions. The current app has no demo-reading mode and creates new readings from its real measurement engine. Maps and comparisons keep warning users when older simulated readings are present.

12. What happens if a test failed at a spot? The value is null and is ignored for that metric. The rest of the reading still counts.

13. What if the user deletes a session? The session and its readings are removed from the phone, after a confirmation.

14. What are the limitations of your analysis?

- Thresholds are untuned starting values.
- Quadrants are coarse: they use the room's bounding box centre.
- Readings are tapped by hand, so positions can be wrong.
- The 4×4 grid is a fixed choice.

15. What would you improve? Tune the thresholds after real-room testing, use finer or adaptive zones, show per-spot detail, and take two tests per spot to reduce noise.

Honest status

- Done: storage, session list, rename and delete, comparison, weak-spot detection, unit tests for the analysis.
- Not done yet: real-room testing in at least three rooms, threshold tuning from that testing, and tuning of the heatmap constants. Say this plainly if asked.